JNTUH Used papers March 2022

Code No: 123AN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, March - 2022
PROBABILITY AND STATISTICS
(Common to ME, CSE, IT, MCT, AME, MIE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) A continuous Random variable has the p.d.f $f(x)=\begin{cases} Kxe^{-\lambda x} & If\ x\geq 0, \lambda \geq 0 \\ 0 & otherwise \end{cases}$
Determine: i) K ii) The mean iii) variance.

b) Assume that 50% of the Engineers are good in Mathematics. Find the probability that among 9:
i) Exactly 5 ii) At least 6 iii) At the most 5 are good in Mathematics. [8+7]

2.a) If the variance of a Poisson variate is 2, find the probability that:
i) P(X = 0) ii) P(1< X < 4) iii) P(0 < X < 3)

b) Suppose the weights of 1000 male students are normally distributed with mean μ =80 with a standard deviation of 10. Find the number of students whose weights are
i) Between 90 and 100 ii) More than 105. [7+8]

3. Calculate the coefficient of correlation and the two lines of regression between the two variables x and y**.** [15]

x	14	16	17	18	19	20	21	22	23
y	84	78	70	75	66	67	62	58	60

4.a) Fit an equation of the form $y = ab^x$ to the following data

x	2	3	4	5	6
y	144	172.8	207.4	248.8	298.5

b) The equations of two regression lines obtained in a correlation analysis are $3x + 12y = 19$, $3y + 9x = 46$ Find:
i) Coefficient of Correlation
ii) The means of x and y
iii) The ratio of the coefficient of variability of x to that of y. [7+8]

5.a) The S.D of a population is 140.5 and the sample size 50. Find the maximum error with 95% confidence.

b) Random samples of 400 men and 600 women in a locality were asked whether they would like to have a bus stop near their residence 200 men and 325 women were in favour of the proposal. Test the significance between the difference of two proportions at 5% level. [7+8]

6. To determine whether really a relationship between an employer's performance in the company in training program and his/her ultimate success in the job taking a sample of 400 cases, from its very extensive files and obtains the results shown in the following table: [15]

	Below Average	Average	Above average
Poor	23	60	29
Average	28	79	60
Very good	9	49	63

7. A P.C repair man finds that the time spent on jobs has an exponential distribution with mean 30 minutes. If the sets are repaired in the order, in which they come in, and if the arrival of sets is approximately Poisson with an average of 10 per 8 hours day. Find:
a) The expected idle time per day
b) The probability that there is no unit in the system.
c) Average number of units in the queue.
d) The average waiting time in queue
e) The probability that a unit will have to wait for more than 10 minutes. [15]

8. In a presidential election at the end of June 40% of the voters registered for liberal, 45% for conservative and 15% for independent. Over one month, of the people those who registered for liberal 80% were retained, 15% changed to conservative and 5% to independent. Of the people those who registered for conservative 70% were retained, 20% changed to liberal and 10% to independent. Of the people those who registered for independent. 60% were retained, 20% changed to liberal and 20% to conservative.
a) Write transition probability matrix
b) Find the % of the voters in each category at the end July and August. [15]

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